REALM WORKSHOP · EMNLP 2026 · BUDAPEST

What Agent Benchmarks Don’t Tell You: Recoverability and Credit Assignment in Agentic Retrieval Evaluation

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THE PROBLEM

Agent quality on long-horizon tasks is reported as **one number**: the aggregate gain of a tool-using agent over a non-agentic baseline. We show this number is **not a property of the agent**.

On the *same* harness, *same* readers, *same* retrieval stack, the agent−baseline delta ranges from **−0.21 to +0.30** across corpora — the sign flips, not just the magnitude.

~6,000	7	3
logged agent episodes	corpora, native metrics	reader families: Claude 4.6, GPT-5.4, Qwen3.6

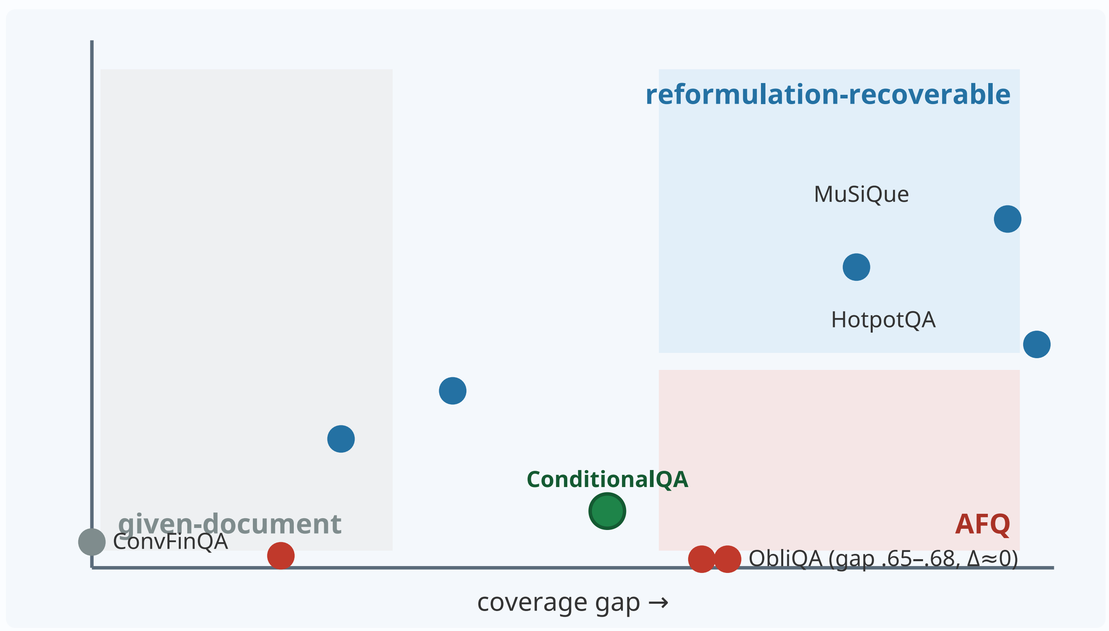
TWO CONFOUNDS ONE NUMBER HIDES

- **Benchmark selection determines the verdict.** Whether extra retrieval can help is a property of the *task*, fixed before any system runs. Subsets with comparable coverage gaps (0.65–0.68 vs ∼1.00) show *opposite* agent value.
- **The gain is credited to the wrong component.** An agent retrieves more than once *and* chooses what to do next. Without a middle arm, evaluations over-credit the expensive component.

1 CHARACTERIZE BEFORE YOU MEASURE

LLM-free **recoverability profile** per benchmark, fixed *before* any outcome:

- **Coverage gap** = 1 − gold-coverage@k
- $\mathbf{R_{ref}}$: misses recoverable by *reformulation* (PRF/Rocchio probe)
- $\mathbf{R_{nav}}$: recoverable by *structural navigation*
- **AFQ**: absent-from-query — addressable by neither



Recoverability plane (schematic). Position — not the gap alone — predicts agent advantage. ObliQA’s high-gap subsets fall *below* the trend: their misses are absent-from-query.

PAPER & CODE

Full paper in the **REALM 2026 proceedings** (ACL Anthology). All corpus builders, characterization probes, scoring scripts, raw generations and grades will be released; ObliQA is additionally graded with a fully local open-model judge so headline scoring reproduces without any metered API.

2 CREDIT ASSIGNMENT BY ABLATION LADDER

Three arms, not two: R1 one-shot RAG → **R2** fixed-iterative loop (no tool choice; an instrument, not a system) → **R3** full agent.

γ = (R2 − R1) / (R3 − R1) = share of the agent’s edge a choice-free system already delivers

Reformulation-recoverable corpora: **γ = 65–96%** under every reader — most of what is filed as evidence for *agency* is evidence for *iteration*. Navigation-recoverable ConditionalQA: γ only 8–41%; R2 saturates by round 2, never reaching the agent.

3 VALIDITY CHECKS THAT FLIP VERDICTS

- Pre-registered **TOST equivalence** makes “no difference” testable (ObliQA strong systems: equivalent)
- **Metric-artifact audit** caught a sign error: naive token-F1 favoured RAG by 0.10; format-aware scoring reversed it
- **Cross-family judge re-grading** changed one magnitude by **3.7×** — report judge magnitudes as judge-conditional

4 PRICE THE HORIZON

Agent runs **1.5–3.6×** RAG’s search calls; fully instrumented: **3.1× latency, 4.4× tokens for zero gain** (ObliQA/GPT). Report the Pareto frontier, not the delta.

HEADLINE RESULTS

Corpus	Regime	Δ	γ	Verdict
ObliQA	AFQ	−0.21…+0.08	—	TOST-equiv. / RAG ahead
HotpotQA	reform.	+0.08…+0.16	65–122%	iterate
MuSiQue	reform.	+0.15…+0.28	58–96%	iterate
ConditionalQA	navig.	+0.16…+0.30	8–41%	**agent earns it (+0.15)**
ConvFinQA·SQuAD·TAT-QA·FinQA	control	0 by constr.	—	no escalation

Not blanket negativity: on navigation-recoverable ConditionalQA, agency keeps a **+0.15 residual under every reader** that no round budget closes.

REPORTING CHECKLIST

- ✓ State each benchmark’s a-priori coverage gap *and* recovery channel; group results by regime.
- ✓ Report three arms, the gap-closing fraction γ, and an R2 round-budget sweep.
- ✓ Test null cells for equivalence; replicate across readers; audit metric artifacts; report judge magnitudes as judge-conditional.
- ✓ Report quality against calls, tokens, and latency.

One-shot RAG when evidence is directly reachable · fixed iteration when misses are reformulation-recoverable · a full agent when misses need structural navigation · no escalation when misses are absent-from-query.